Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1(Adv-2019-P1-Model)** **Date: 11-02-2026**

Time: 09.00Am to 12.00Pm **GTA-16** **Max. Marks: 180**

KEY SHEET

PHYSICS

1	**C**	2	**A**	3	**C**	4	**A**	5	**ABC**	6	**AB**
7	**ACD**	8	**AC**	9	**AC**	10	**BD**	11	**AC**	12	**AC**
13	**6**	14	**2**	15	**6**	16	**2**	17	**3.64**	18	**0.30**

CHEMISTRY

19	**D**	20	**B**	21	**C**	22	**A**	23	**AB**	24	**ABD**
25	**ABC**	26	**BCD**	27	**AC**	28	**AD**	29	**ABCD**	30	**ABC**
31	**0**	32	**6**	33	**8**	34	**4**	35	**2**	36	**8**

MATHEMATICS

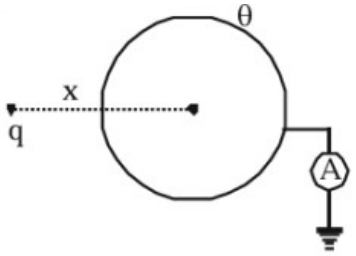
37	**C**	38	**C**	39	**A**	40	**D**	41	**AD**	42	**BCD**
43	**BCD**	44	**ACD**	45	**BCD**	46	**ABD**	47	**B**	48	**BC**
49	**1**	50	**12**	51	**7**	52	**9**	53	**5**	54	**4**

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SOLUTIONS

PHYSICS

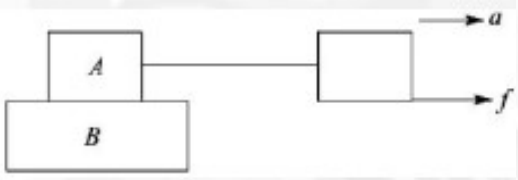
1. $\frac{kq}{x}+\frac{kQ}{b}=0$



$\theta=\frac{qb}{x}i=\frac{dQ}{dt}=\frac{-qb(-V)}{x^2}=\frac{qbV}{x^2}$

2. In the given situation potential difference is independent of charge on outer shell.

3. If all blocks moves with same acceleration



$a=\frac{F}{2M+2m}$

$F\le\frac{2\mu smg(m+M)}{2m+M}$ $\quad f=(m+m+M)a=\frac{(2m+M)F}{2(m+M)}$

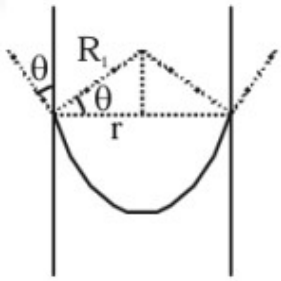
For no slipping $f\le f_l$ $\quad \frac{(2m+M)}{2(m+M)}F\le\mu smg$

4. $KE_{loss}=\frac{1}{2}\frac{m\times 7m}{8m}(u-0)^2(1-e^2)$

$KE_{min}\Rightarrow u_{min}\Rightarrow e=0$ $\quad \Rightarrow\left(\frac{1}{2}mu_{min}^2\right)\times\frac{7}{8}=KE_{loss}$

also $\frac{7}{8}K_{min}+Q=0$ $\quad \Rightarrow K_{min}=-\frac{8}{7}Q=1880\text{ keV}$

5. Fig. a and b



$\cos\theta=\frac{r}{R_1}\Rightarrow R_1=\frac{d}{2\cos\theta}$ $\quad \Delta P=S\left(\frac{1}{r/\cos\theta}+\frac{1}{\infty}\right)=\frac{S\cos\theta}{r}=\frac{2S\cos\theta}{d}$

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$$\Rightarrow \Delta P = h\rho g = \frac{2S\cos\theta}{d} \Rightarrow h = \frac{2S\,cso\theta}{\rho g d}$$

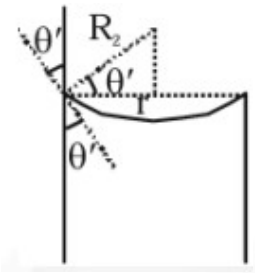
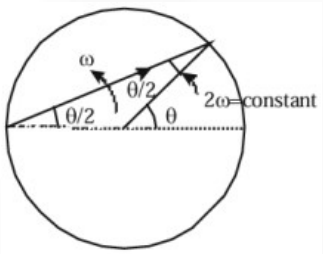


Fig (c) : $\cos\theta' = \frac{r}{R_2} \Rightarrow R_2 = \frac{r}{\cos\theta'} = \frac{d}{2\cos\theta'}$

as $\Delta P = \frac{\rho g h}{2} = \frac{2S\cos\theta'}{d} \Rightarrow \cos\theta' = \frac{\cos\theta}{2}$

6.



It is uniform circular motion with angular speed = 2ω about centre.

Acceleration is variable in direction, but constant in magnitude = $4\omega^2 R$

7. $U = 3x + 4y$ $\quad \vec{F} = (-3)\hat{i} + (-4)\hat{j} = m\vec{a} \Rightarrow \vec{a}$ = constant

$$\vec{r} = \left(6 - \frac{3}{2}t^2\right)\hat{i} + \left(4 - 2t^2\right)\hat{j}$$

8. $m\vec{v}_B + 2m\vec{v}_{AC} = 0$ $\qquad \vec{v}_{AC} = \frac{-\vec{v}_B}{2}$

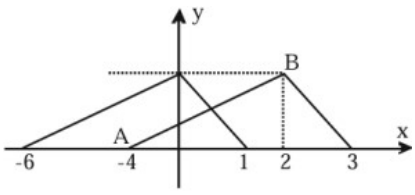
9. $A_1 v_1 = A_2 v_2$

$$W = (0.1)(10)(0.05) + (0.02)(10)\left(\frac{0.05}{2}\right) + \frac{1}{2}\left(\frac{20}{1000}\right)(2)^2 \quad W = 0.095$$

10. $204 = -13.6\, z^2\left(\frac{1}{1} - \frac{1}{4n^2}\right)$ $\qquad 40.8 = -13.6\, z^2\left(\frac{1}{n^2} - \frac{1}{4n^2}\right)$

$$5 = \frac{1 - \frac{1}{4n^2}}{\frac{1}{n^2} - \frac{1}{4n^2}} \Rightarrow n = 2 \Rightarrow z = 4 \qquad E_1 = -13.6\frac{(16)}{(1)^2} = -217.6\ eV$$

11.



For AB : At $t = 0.2s$

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$$y-1=\frac{1}{6}(x-2)\Rightarrow y=\frac{x}{6}+\frac{2}{3} \qquad x=1\text{ cm} \Rightarrow y=\frac{5}{6}\text{cm}$$

$$\text{Transverse velocity} = \text{-slope} \times V_w \qquad =\left(-\frac{1}{6}\right)\times 10=\frac{5}{3}\text{ cm/s}$$

12. Let angular velocity of centre of ring about vertical axis passing through center of circular path is ω_2

$$\text{Then : } \omega_2 R=\omega r \qquad \omega_2=\frac{\omega r}{R} \qquad \vec{\alpha}=\vec{\omega}_2\times\vec{\omega}\Rightarrow\alpha=\omega_2\omega=\frac{\omega^2 r}{R}$$

13. $$W_g=\frac{1}{2}kx^2+P_0Ax=\frac{1}{2}\times 8000\times(0.1)^2+10^5\times 8\times 10^{-3}\times 0.1=40+80=120\text{ J}$$

$$\text{also } P_f=P_0+\frac{Kx}{A}=10^5+\frac{8000\times 0.1}{8\times 10^{-3}}=2\times 10^5\text{ Pa} \because \frac{P_f\ V_f}{P_i\ V_i}=\frac{T_f}{T_i}$$

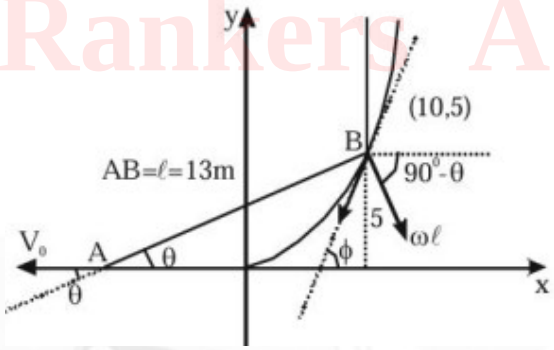
$$=T_f=\frac{2\times 10^5\times 3.2\times 10^{-3}}{10^5\times 2.4\times 10^{-3}}\times 300=800\text{K}$$

$$\text{Heat supplied by heater} = nC_v\Delta T+W_g$$

$$=\frac{10^5\times 2.4\times 10^{-3}}{R\times 300}\times\frac{3R}{2}\times(500)+120=600+120=720\text{J}$$

$$\text{Required ration} = \frac{720}{120}=6$$

14.

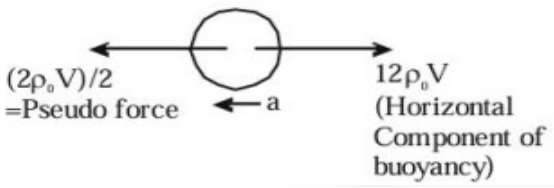


$$\tan\phi=\frac{dy}{dx}=\frac{2x}{20}=\frac{2\times 10}{20}=1 \qquad \phi=45^0$$

$$\vec{v}_B=\vec{v}_A+\vec{\omega}\times\overrightarrow{AB}\Rightarrow\frac{v_B}{\sqrt{2}}\left(-\hat{i}-\hat{j}\right) \qquad =34\left(-\hat{i}\right)+\omega\times 13\left\{\sin\theta\hat{i}+\cos\theta\left(-\hat{j}\right)\right\}$$

$$\Rightarrow\frac{v_B}{\sqrt{2}}=34-5\omega \text{ and } \frac{v_B}{\sqrt{2}}=12\omega \qquad \Rightarrow 34-5\omega=12\omega\Rightarrow\omega=2\text{ rad/s}$$

15.



$$a=\frac{12\rho_0 V}{2\rho_0 V}=6\text{ m/s}^2$$

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16. $\frac{5\lambda}{4}=l \Rightarrow \lambda=\frac{4l}{5} \quad \frac{\lambda}{8}=\frac{l}{10} \Rightarrow \phi=\frac{2\pi}{\lambda}\times\frac{\lambda}{8}=\frac{\pi}{4}$

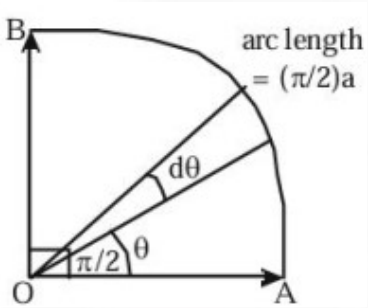
$\text{amp}=\text{A}\ \sin(\text{kx})=\text{A}\sin\left(\frac{\pi}{4}\right)=\frac{\text{A}}{\sqrt{2}}\Rightarrow\alpha=2$

17. $\text{L.C}=\frac{1\text{mm}}{50}$ zero error $=-6\times\frac{1\text{mm}}{50}$

reading $=3\text{ mm}+26\times\text{L.C}-\text{zero error}$

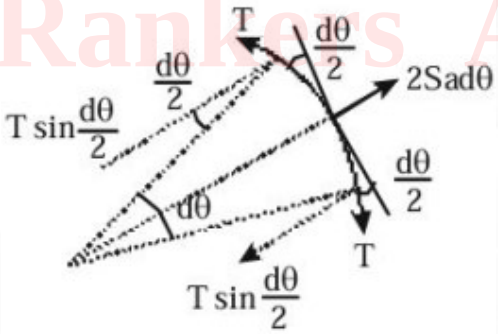
$=3\text{mm}+26\times\frac{1\text{mm}}{50}-\left(-6\times\frac{1}{50}\text{mm}\right)=3.64\text{ mm}$

18.



OA = OB = a = distance from circumference

Let us take as a radius of the arc then length should be $\frac{\pi}{2}$a, which is satisfied. Hence 'a' is radius of quarter arc.



$\Rightarrow 2\text{T}\sin\left(\frac{\text{d}\theta}{2}\right)=2\text{sad}\,\theta$

$\Rightarrow \text{Td}\theta=2\text{sad}\theta$

$\Rightarrow \text{T}=2\text{ Sa}=2\times3\times10^{-3}\times5\times10^{-2}=0.30\text{ mN}$

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CHEMISTRY

19. Compound P is polymer of 2-methylbuta-1, 3-diene

20. EAN of $K_3[Fe(CN)_6] = 35$, $[Co(en)_3]Cl_3$ can show optical isomerism

21. $Be_{(g)} + e^- \rightarrow Be^-_{(g)}$ is an endothermic process So $Be^-_{(g)}$ is unstable.

22. a = 356 pm $\quad 2r = \frac{a\sqrt{3}}{4} \quad r = \frac{a\sqrt{3}}{8} \quad r = 77.07$ pm

23. Conceptual

24. Conceptual

25. Peptide contains both sulphur and nitrogen therefore it gives bloodred colour with $FeCl_3$ in Lassaigne's test, On hydrolysis of above peptide gives basic amino acid i.e lysine

26. (I) $Th : [Rn]6d^2 7s^2$

(II) C – O bond length : maximum in $[V(CO)_6]^-$ due to greater extent of synergic bonding

(III) MnO_4^- ; diamagnetic and MnO_4^{2-} ; paramagnetic

27. (I) $Fe_2(SO_4)_3 \xrightarrow{\Delta} Fe_2O_3 + SO_3$ $\quad$ (II) $FeSO_4 \xrightarrow{\Delta} Fe_2O_3 + SO_3 + SO_2$

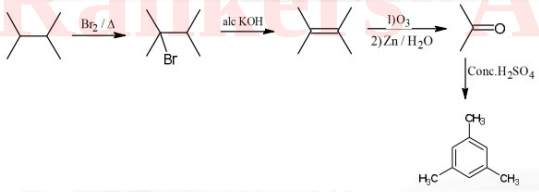
(III) $Ag^+ \xrightarrow{CN^-} AgCN\downarrow(\text{white}) \xrightarrow{CN^-} \text{soluble } [Ag(CN)_2]^-$

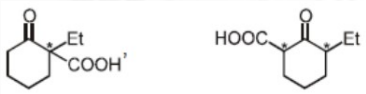
$Pb^{+2} \xrightarrow{CN^-} Pb(CN)_2\downarrow(\text{white}) \xrightarrow{CN^-} \text{insoluble}$

28. Conceptual

29. Conceptual

30. Conceptual

31. 

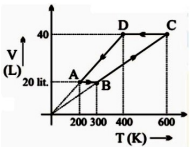
32. 

33. $n_{eq} Fe_2(C_2O_4)_3 + n_{eq} FeC_2O_4 = n_{eq} KMnO_4$

or, $4.5 \times 6 + 4.5 \times 3 = n \times 5 \qquad \therefore n = 8.1$

34. ZnS – white, NiS – black

35. $PV = RT + Pb \quad V = \frac{RT}{P} + b \quad \text{slope} = \frac{R}{P} = \frac{0.0821}{04105} = 2$

36. 

$W_{AB} = W_{CD} = 0$

$W_{BC} = -nR\Delta T = -1 \times R \times 300 = 300R$

$W_{DA} = -nR\Delta T = -1 \times R \times -200 = 200R$

$W_{total} = -100R \qquad \Rightarrow |W| = 8$ atm. lit

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MATHEMATICS

37. $f'(x) = x^2 + ax + b$ is injective if $D \le 0$

$a^2 - 4b \le 0$

If a = 1, b = 1, 2, 3, 4, 5 Number of pair = 5
a = 2, b = 1, 2, 3, 4, 5 Number of pair = 5
a = 3, b = 3, 4, 5 Number of pair = 3
a = 4, b = 4, 5 Number of pair = 2
a = 5 b has no value

38. Required probability = $\frac{3! \times 2}{9!} = \frac{1}{140}$ 3!3!3! 3!(3n – 3)!

39. $\frac{x}{5}$ is integer at 21 points in [0, 100]

$\frac{x}{2}$ is integer at 51 points in [0, 100]

$\therefore$ but when 'x' is a multiple of 10 then f(x) is continuous.

So that respective points should be subtract from both i.e., multiple of 10 are 11 points in [0, 100] 21 + 51 – 11 – 11 = 72 – 22 = 50

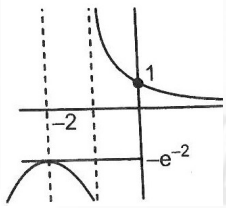
40. $\left|\text{adj}\left(\frac{M}{2}\right)\right| = \left|\frac{M}{2}\right|^2 = \left(\frac{1}{8}|M|\right)^2$

41. $L_1 : x = y = z$ $L_2 : \frac{x-1}{1} = \frac{y+1}{-1} = \frac{z}{-1}$, Shortest distance = $\frac{1}{\sqrt{2}}$

Equation of plane containing line L_2 and parallel to L_1 $y - z + 1 = 0$

Distance of origin from this plane = $\frac{1}{\sqrt{2}}$

42. $f(x) = \frac{e^{-x}}{1+x}$



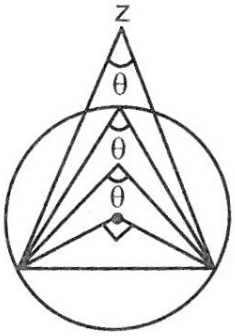
$f'(x) = \frac{-e^{-x}(x+2)}{(1+x)^2}$

43. A) If $\arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{2}$ then z_1 and z_2 subtend right-angle at circumcentre origin.

$\therefore$ the chord joining z_1 and z_2 will subtend an angle θ at 'z' such that

$$\begin{cases} \theta = \frac{\pi}{4} & \text{if} \quad |z| = 1 \\ \theta < \frac{\pi}{4} & \text{if} \quad |z| > 1 \\ \theta > \frac{\pi}{4} & \text{if} \quad |z| < 1 \end{cases}$$

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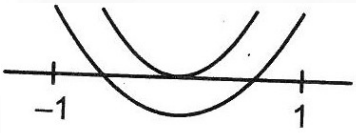


B) $|z_1z_2+z_2z_3+z_3z_1|=|z_1|.|z_2|.|z_3|\left|\frac{1}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}\right| \quad =\left|\overline{z_1+z_2+z_3}\right|$

C) $\left(\frac{(z_1+z_2)(z_2+z_3)(z_3+z_1)}{z_1.z_2.z_3}\right)=\left(\frac{\overline{(z_1+z_2)(z_2+z_3)(z_3+z_1)}}{z_1.z_2.z_3}\right)$

D) The triangle formed by joining z_1, z_3 and z_2 is isosceles and right angled at z_3

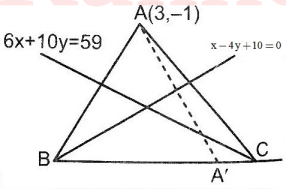
44. $D\geq 0\cap f(-1)>0\cap f(1)>0$



$-2<K\leq\frac{1}{4}$

45. Image of A(3, -1) about angle bisector x – 4y + 10 = 0 is $A'(a, b)$

$\frac{a-3}{1}=\frac{b+1}{-4}=\frac{-2(3+4+10)}{17}$



$\Rightarrow A'(1, 7)$ Let point $B\left(x_1, \frac{x_1+10}{4}\right)$ on the line x – 4y + 10 = 0

If mid point of AB lie on the line 6x + 10y = 59

$6\left(\frac{x_1+3}{2}\right)+10\left(\frac{x_1+10-4}{8}\right)=59 \quad \Rightarrow B(10, 5)$

46. $\sum_{k=0}^{100}{}^{100}C_k(x-2)^{100-k}.3^k=(x+1)^{100}$ coeff. of $x^{50}={}^{100}C_{50}$

47. $\int\frac{(1+\sin^{2015}x)-\sqrt{1+\sin^{4030}x}}{2\sin^{2015}x}$ (Rationalise) $\int_{-2014}^{2014}\frac{1}{2}dx \quad \int odd=0$

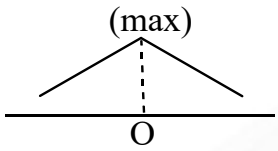
48. $\frac{10^{n+1}+1}{10^{n+2}+1}<\frac{10^{m+1}+1}{10^{m+2}+1}$

$\Rightarrow 10^{n+1}.10^{m+2}+10^{n+1}+10^{m+2}+1<10^{n+2}.10^{m+1}+10^{n+2}+10^{m+1}+1 \quad \Rightarrow 10^{m+1}<10^{n+1}$

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49. Put $j=0, S_0=\dfrac{1+e^{-\pi}}{2}$

50. $I(t)=\int_{\alpha}^{\beta}\left(x^2+2x-t^2\right)dx=\dfrac{x^3}{3}+x^2-t^2\ x\Big|_{\alpha}^{\beta}$



$I(t)=\dfrac{\beta^3-\alpha^3}{3}+\left(\beta^2-\alpha^2\right)-t^2(\beta-\alpha)\quad I'(t)=-2t(\beta-\alpha)=0$

$I(t)\le I(0)\qquad \therefore I(0)=\int_{-2}^{0}\left(x^2+2x\right)dx=-\dfrac{4}{3}\Rightarrow\dfrac{p}{q}=|I(0)|=\dfrac{4}{3}$

51. Equation of transverse axis is given by $\dfrac{|3x-4y-1|}{\sqrt{3^2+(-4)^2}}=\dfrac{|4x-3y-6|}{\sqrt{(4)^2+(-3)^2}}$

And $x-y-7=0$ $\therefore$ Transverse axis is given

52. $5-2\pi>x^2-4x\quad x^2-4x+(2\pi-5)<0$

$2-\sqrt{9-2\pi}<x<2+\sqrt{9-2\pi}\Rightarrow\lambda=9$

53. $\sigma^2=\dfrac{n_1\left(d_1^2+\sigma_1^2\right)+n_2\left(d_2^2+\sigma_2^2\right)}{n_1+n_2}$

$A=\dfrac{n_1\bar{x}+n_2\bar{y}}{n_1+n_2}=\dfrac{10.2+n.3}{n+10}=\dfrac{3n+20}{n+10}$

$d_1=\bar{x}-A=2-\dfrac{3n+20}{n+10}=\dfrac{2n+20-3n-20}{n+10}=\dfrac{-n}{n+10}$

$\dfrac{17}{9}=\dfrac{10\left(n^2+2(n+10)^2\right)+n\left(100+n(n+10)^2\right)}{(n+10)^3}$

$10n+20n+200=\dfrac{n^2-10n}{(n+10)^2}=\dfrac{17}{9}$

$=17n^2=340n+1700$

$8n^2-20n-100=0\qquad 2n^2-5n-25=0$

$(2n-10)(n+5)=0\quad n=5$

54. $4\in A$ but not in $C|A|.\ |B|=\ |A\cap B|.|A\cup B|\Rightarrow A\subseteq B$ or $B\subseteq A$ number of ordered pair (A, B) is

$A\subseteq B\ \ {}^5C_0.2^5+{}^5C_1.2^4+{}^5C_2.2^3+{}^5C_3.2^2+{}^5C_4.2+{}^5C_5$

$=(1+2)^5=3^5\ \ R.V=2.3^5-2^5=454=2\times227=4$

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